

# Curriculum Vitae

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**Name:** Sebastian G. Gruber

**E-Mail:** sebggruber@gmail.com

**Website:** <https://sebggruber.github.io/>

**Research Interests:** Uncertainty Quantification (especially Calibration), Proper Scoring Rules, Probability Theory, Kernel Methods, Generative Model Evaluation

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## Grants

**Receiver of German Cancer Research Center**

**Short-term Travel Grant**

Mar. 2024 - May 2024

*Total worth: 1850 Euros*

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## Education

**Goethe University Frankfurt**

Mar. 2021 - Jan. 2025

*Computer Science PhD, Grade: –*

- Thesis in machine learning with title: “A Novel Framework for Uncertainty Quantification via Proper Scores for Classification and Beyond”
- Funded by ERC Consolidator Grant (Project: TAIPO)

**Ludwig Maximilian University Munich**

Oct. 2018 - Feb. 2021

*Statistics MSc, Grade: 1.4 (Very good)*

Thesis in machine learning about meta-learning and knowledge distillation

**Ludwig Maximilian University Munich**

Oct. 2015 - Sep. 2018

*Statistics BSc, Grade: 1.43 (Very good)*

Thesis in machine learning about reinforcement learning

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## Work Experience

**KU Leuven (Leuven and Remote)**

May 2025 - Today

*Postdoctoral Researcher in PSI division at ESAT department*

**German Cancer Research Center (Frankfurt)**

Feb. 2025 - Apr. 2025

*Postdoctoral Researcher in Machine Learning for Oncology Group*

**Inria (Paris)**

Mar. 2024 - May 2024

*Doctoral Visiting Student in Sierra Team*

**German Cancer Research Center (Frankfurt)**

Mar. 2021 - Jan. 2025

*Doctoral Student in Machine Learning for Oncology Group*

Research on uncertainty quantification for trustworthy AI

**Siemens (Munich)**

Aug. 2018 - Feb. 2021

*Working/Master Student in Machine Learning*

- Research followed by Master thesis at the Machine Intelligence Research Group of the Corporate Technology department
- Machine Learning end-to-end service at the Analytics Lab of the IT department

**Criteo (Munich)**

July 2017 - July 2018

*Working Student Data Analytics*

## Open Source Projects

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**mlr3hyperband CRAN Package**

2019/2020

*R package for hyperparameter optimization in the mlr3 ecosystem*

<https://github.com/mlr-org/mlr3hyperband>

**mlrPlayground Web App**

2018/2019

*Web application showcasing Machine Learning algorithms for educational purposes*

<https://sebastian-gruber.shinyapps.io/mlrPlayground/>

## Selected Publications

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**Gruber, S. G.**, Bach, F. (2025) *Optimizing Estimators of Squared Calibration Errors in Classification*. In Transactions on Machine Learning Research.

**Gruber, S. G.**, Ziegler, P. T., Buettner, F. (2024) *Disentangling Mean Embeddings for Better Diagnostics of Image Generators*. In Interpretable AI: Past, Present and Future Workshop @ NeurIPS.

**Gruber, S. G.**, & Buettner, F. (2024). *A Bias-Variance-Covariance Decomposition of Kernel Scores for Generative Models*. In International Conference on Machine Learning, 41, (pp. 16460-16501).

**Gruber, S. G.**, Popordanoska, T., Tiulpin, A., Buettner, F., & Blaschko, M. B. (2024). *Consistent and Asymptotically Unbiased Estimation of Proper Calibration Errors*. In International Conference on Artificial Intelligence and Statistics, 27, (pp. 3466-3474).

**Gruber, S. G.**, & Buettner, F. (2023). *Uncertainty Estimates of Predictions via a General Bias-Variance Decomposition*. In International Conference on Artificial Intelligence and Statistics, 26, (pp. 11331-11354).

**Gruber, S. G.**, & Buettner, F. (2022). *Better uncertainty calibration via proper scores for classification and beyond*. In Advances in Neural Information Processing Systems, 35, 8618-8632.

*In total 9 publications and 1 patent*

(c.f. [https://scholar.google.com/citations?user=\\_ThqALUAAAAJ](https://scholar.google.com/citations?user=_ThqALUAAAAJ))

## Conference Talks

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Trustworthy AI for Medical Image Analysis and Computer Vision Workshop (2025): *“Uncertainty Calibration via Proper Scores for Trustworthy AI”*

Helmholtz AI Conference (2024): *“A Kernel-based Framework for Uncertainty in Generative AI”*

AI InScideOut Unconference (2023): *“Enhancing the Reliability of Large Language Models: A Novel Theoretical Framework for Uncertainty Assessment”*

## Invited Talks

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Trustworthy AI for Medical Image Analysis and Computer Vision workshop (2025): *“Uncertainty Calibration via Proper Scores for Trustworthy AI”*

Inria Saclay Soda Group (2024): *“Proper Calibration Errors for Classification and Generative AI”*

Inria Paris Sierra Group (2024): *“A Framework for Uncertainty Estimation in Machine Learning based on Proper Scores”*

LMU Munich Statistics Department (2023): *“Uncertainty Estimates of Predictions via a General Bias-Variance Decomposition”*

Bayer AG Research Department (2022): *“Trustworthy Machine Learning in Oncology”*

Schulz Lab (2022): *“Trustworthy Deep Learning via Proper Calibration Errors: A Unifying Approach for Quantifying the Reliability of Predictive Uncertainty”*

## Teaching and Supervision

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Co-supervised one master thesis and 7 bachelor theses in computer science at Goethe University.

Gave multiple lectures in the course “Introduction to Methods of Artificial Intelligence” at Goethe University.

Organised the tutorials and designed the exams of the course “Introduction to Methods of Artificial Intelligence” at Goethe University in summer 2021, summer 2022, and summer 2023.

## Skills

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*Languages*            German (native), English (fluid), French (basic)

*Programming Languages*    Python, R, Julia, SQL, Haskell